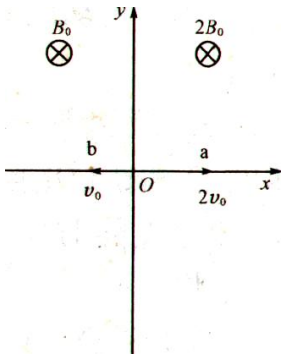
point inward. At O, a positive particle has charge q and mass m. Particle a at $t=0$ has speed $2v_0$ along positive x. After a starts, Like a in mass and charge, b leaves O at v_0 along negative x axis. For a and b to meet, find the least difference between their launch times and the corresponding meeting coordinates. Assume the field is vacuum; ignore gravity and interaction between a、b.



5.

Unaided, a myopic eye sees clearly only within $a=20\text{cm}$. Holding a card near the eye, he sees distant objects more clearly through its small circular hole, without glasses.

1. A hole has diameter D. Find the retinal spot diameter for a distant point source, viewed directly by the myopic eye, using geometrical optics.

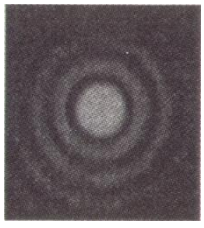
2. Include diffraction; find optimum hole diameter. Use mean visible wavelength 600nm。

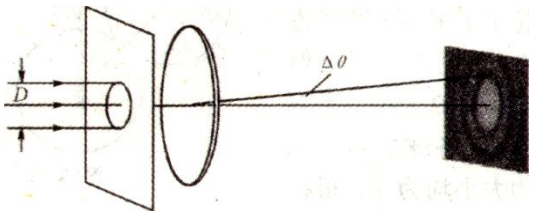
Hints: 1. Model the complex eye as a thin lens of adjustable focal length f and an imaging retina. The lens rim is the pupil; air on both sides, retina-lens spacing b。

2. The hole's effect: geometrical optics focuses distant-point light parallel to the axis at the lens focus. A circular aperture of diameter D placed before the lens diffracts light. At the focus the pattern shown appears: a bright central disk surrounded by fading concentric bright/dark rings. Being much fainter, these may be ignored. The disk's angular radius $\Delta\theta$ at the lens center depends on D. Theory gives

$$\Delta\theta = 1.22\frac{\lambda}{D}$$

where λ is wavelength. The shown disk is the point image through the hole. Larger D gives a smaller, sharper, more point-like image. Smaller D gives a larger diffraction spot and a image less clear. Near the focus the screen shows a similar pattern, and the central disk's angular radius $\Delta\theta$ is approximately given by the same formula.





6.

Spherical Earth has radius 6370km. In the equatorial annulus to over 1 thousand km altitude, geomagnetic field of flux density $B=3.20\times10^{-6}\text{T}$. A charged cosmic particle has rest mass $m_0=6.68\times10^{-27}\text{kg}$; charge $q=3.20\times10^{-19}\text{C}$. In the uniform equatorial field, orbit radius is $R=7370\text{km}$. The circle is centered on Earth.

In relativity, when force F always acts perpendicular to velocity v, mass m and acceleration a still relate to force F by $F=ma$, but m is the relativistic mass. Find: